

Blue_Observer

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Site d'observation de biodiversité marine

<http://127.0.0.1:5000/>



UNIVERSITÉ de
MONTPELLIER

Développement logiciel 2025

Écotourisme et curiosité scientifique

Présentation du sujet

Objectifs principal du projet

Sources de données (OBIS, Copernicus, Open-Meteo)

Problématique

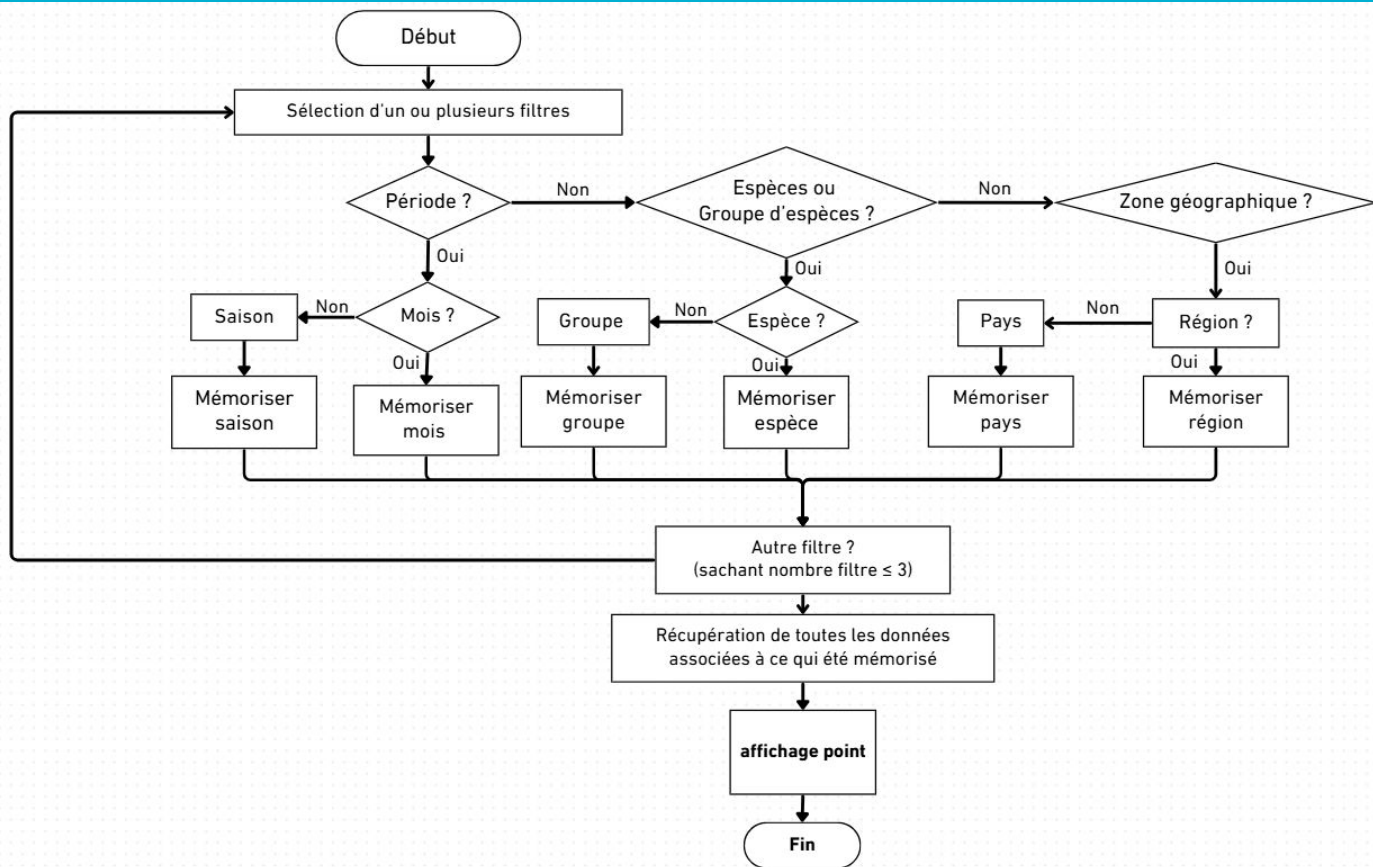


<https://www.ladepeche.fr/article/2010/03/10/794020-mexique-pachico-invente-nouveau-tourisme-caressant-baleine.html>



<https://www.guide-du-paysbasque.com/fr/tourisme/se-divertir/nautiques/bateaux/hendaia-66/explore-ocean-2931.html>

Pipeline



Copernicus : Téléchargement direct

The screenshot displays the homepage of the Copernicus Marine Service website. The browser's address bar shows the URL `marine.copernicus.eu`. The top navigation bar includes links for Resources, News, Press, Events, Contact, a REGISTER button, a search bar, and a language selector set to English. Below this, a secondary navigation bar features links for Services, Opportunities, Access Data, Use Cases, User Corner, and About. The main header area contains the European Union flag, the Copernicus logo (with the tagline 'Europe's eyes on Earth'), and the Copernicus Marine Service logo. The central banner features the text 'Copernicus Marine Service' and a description: 'Providing free and open marine data and services to enable marine policy implementation, support Blue growth and scientific innovation.' Below the banner, an 'Access Data' link is followed by four service categories: DATA (OCEAN PRODUCTS), EXPERTISE (OCEAN STATE REPORT), TRENDS (OCEAN CLIMATE TRENDS), and EXPLORATION (OCEAN VISUALISATION). Each category includes a brief description and links to further resources. The background of the page is a high-resolution image of ocean waves crashing against dark rocks.

← → ↻ 🔍 marine.copernicus.eu

Resources News Press Events Contact **REGISTER** Type... English

Services Opportunities Access Data Use Cases User Corner About

Copernicus Marine Service

Providing free and open marine data and services to enable marine policy implementation, support Blue growth and scientific innovation.

[Access Data >](#)

DATA	EXPERTISE	TRENDS	EXPLORATION
OCEAN PRODUCTS A robust ocean data catalogue, to download or visualise data including hindcasts, nowcasts and forecasts.	OCEAN STATE REPORT Extensive annual analysis on the state of the ocean over nearly 20 years and severe/notable annual events.	OCEAN CLIMATE TRENDS Monitoring the health of the ocean. Ocean Monitoring Indicators Ocean Climate Portal	OCEAN VISUALISATION Dive into our 4D digital oceans through our 3 visualisation tools for beginner, intermediate and advanced users

Copernicus : Téléchargement direct

▼ Main filters ^

🔍 Free text or ID

Variables

Wave significant height X

Temperature X

Pick one or more

Areas

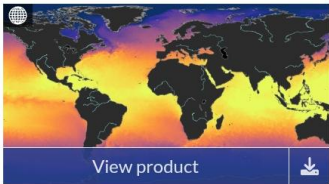
Pick one or more

Time range

← DD/MM/YYYY

→ DD/MM/YYYY

★ Favourites ▼



View product

Global Ocean Physics Analysis and Forecast

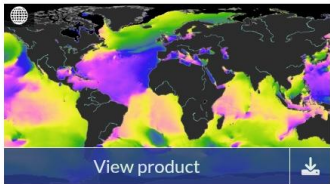
GLOBAL_ANALYSISFORECAST_PHY_001_024

Models

Global, 0.083° × 0.083° × 50 levels

1 Nov 2020 to 16 Dec 2025, hourly, daily, monthly

Temperature, wave, salinity, sea surface height, velocity, mixed layer thickness...



View product

Global Ocean Waves Analysis and Forecast

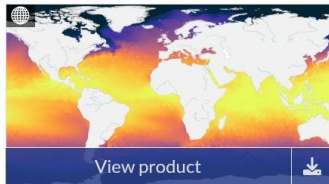
GLOBAL_ANALYSISFORECAST_WAV_001_027

Models

Global, 0.083° × 0.083°

1 Nov 2022 to 16 Dec 2025, hourly

Wave, velocity



View product

Global Ocean Physics Reanalysis

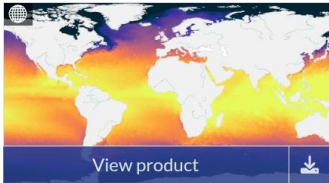
GLOBAL_MULTIYEAR_PHY_001_030

Models

Global, 0.083° × 0.083° × 50 levels

1 Jan 1993 to 28 Oct 2025, daily, monthly

Temperature, salinity, sea surface height, velocity, mixed layer thickness, sea ice



View product

Global Ocean Ensemble Physics Reanalysis

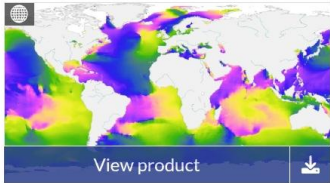
GLOBAL_MULTIYEAR_PHY_ENS_001_031

Models

Global, 0.25° × 0.25° × 75 levels

1 Jan 1993 to 31 Dec 2023, daily, monthly

Temperature, salinity, sea surface height, velocity, mixed layer thickness, sea ice



View product

Global Ocean Waves Reanalysis

GLOBAL_MULTIYEAR_WAV_001_032

Models

Global, 0.2° × 0.2°

1 Jan 1980 to 30 Sep 2025, hourly, monthly

Wave, velocity



View product

Global ocean low and mid trophic levels biomass content hindcast

GLOBAL_MULTIYEAR_BGC_001_033

Models

Global, 0.083° × 0.083° × 3 levels

1 Jan 1998 to 31 Dec 2024, daily

Temperature, velocity, nekton, plankton, optics



Copernicus : Téléchargement direct

The screenshot shows a web browser window with the URL `data.marine.copernicus.eu/product/GLOBAL_MULTIYEAR_PHY_ENS_001_031/download`. The page has a navigation bar with buttons for 'Download', 'Automate', and 'Browse files'. A file size indicator shows '~ 94.15 MB'. The 'Dataset' section displays the product identifier 'GLOBAL_MULTIYEAR_PHY_ENS_001_031', the product name 'Global Ocean Ensemble Physics Reanalysis', and a dropdown menu for the dataset, currently set to 'All reanalyses, daily · cmems_mod_glo_phy-all_my_0.25deg_P1D-m'. The 'Variables*' section includes 'Add all' and 'Clear all' buttons, followed by two columns of variables, each with a checked checkbox and its name and units.

Dataset

Product identifier: GLOBAL_MULTIYEAR_PHY_ENS_001_031
Product name: Global Ocean Ensemble Physics Reanalysis
Dataset: Please choose one of the datasets in this product:
All reanalyses, daily · cmems_mod_glo_phy-all_my_0.25deg_P1D-m

Variables* Add all Clear all

☒ Eastward sea water velocity <code>uo_cglo</code> [m/s]	☒ Sea ice thickness <code>sithick_cglo</code> [m]
☒ Eastward sea water velocity <code>uo_glor</code> [m/s]	☒ Sea ice thickness <code>sithick_glor</code> [m]
☒ Eastward sea water velocity <code>uo_oras</code> [m/s]	☒ Sea ice thickness <code>sithick_oras</code> [m]
☒ Northward sea water velocity <code>vo_cglo</code> [m/s]	☒ Sea surface height above geoid <code>zos_cglo</code> [m]
☒ Northward sea water velocity <code>vo_glor</code> [m/s]	☒ Sea surface height above geoid <code>zos_glor</code> [m]
☒ Northward sea water velocity <code>vo_oras</code> [m/s]	☒ Sea surface height above geoid <code>zos_oras</code> [m]
☒ Ocean mixed layer thickness defined by sigma theta <code>mlotst_cglo</code> [m]	☒ Sea water potential temperature <code>thetao_cglo</code> [°C]
☒ Ocean mixed layer thickness defined by sigma theta <code>mlotst_glor</code> [m]	☒ Sea water potential temperature <code>thetao_glor</code> [°C]
☒ Ocean mixed layer thickness defined by sigma theta <code>mlotst_oras</code> [m]	☒ Sea water potential temperature <code>thetao_oras</code> [°C]
☒ Sea ice area fraction <code>siconc_cglo</code>	☒ Sea water salinity <code>so_cglo</code> [$/ 10^3$]
☒ Sea ice area fraction <code>siconc_glor</code>	☒ Sea water salinity <code>so_glor</code> [$/ 10^3$]
☒ Sea ice area fraction <code>siconc_oras</code>	☒ Sea water salinity <code>so_oras</code> [$/ 10^3$]

OBIS : Téléchargement direct

The screenshot shows the OBIS Mapper web application in a browser. The address bar displays 'mapper.obis.org'. The main area is a world map with navigation controls (zoom in, zoom out, home) on the left. On the right, a sidebar contains a 'Create layer' section with a 'Save' button, a 'Scientific name' input field, and a table for 'Scientific name' and 'Taxon ID'. Below this is a 'Datasets' section with a list of categories: Nodes, Institutes, Areas, Publisher countries, Geometry, Time range, and Depth range. The 'Time range' section shows a red timeline from 1900 to 2025.

← → ↻ mapper.obis.org

+

-

🏠

Create layer Save

Scientific name

Enter scientific name

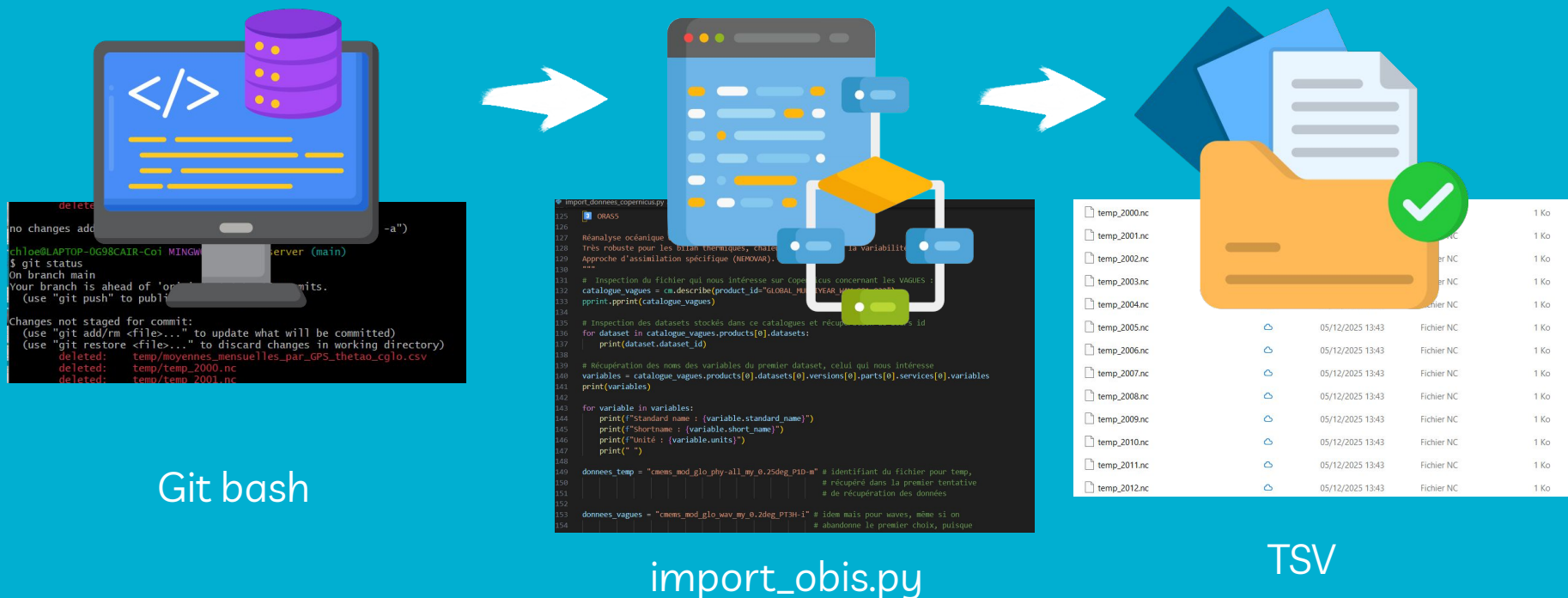
Scientific name	Taxon ID
No taxa selected.	

Datasets

- Nodes
- Institutes
- Areas
- Publisher countries
- Geometry
- Time range
- Depth range

1900 2025

OBIS : Programmation de la récupération via API



OpenMeteo : Téléchargement direct



Weather Forecast

Historical Weather

Ensemble Models

Seasonal Forecast

Climate Change

Marine Forecast

Air Quality

Satellite Radiation

Now, with the addition of the 9-kilometre ECMWF IFS model, the historical weather API provides access to a staggering 90 terabytes of meteorological data! Read the [blog article](#).

Location and Time

Location: ☒ Coordinates ☐ List ☐ Bounding box

Latitude North
69

Longitude West
-27

Longitude East
47

Using bounding box will select grid cells within a specified area, excluding the top and right boundaries. For example, a latitude range of $40^{\circ} \dots < 45^{\circ}$ will not include any grid points on the 45° line. This exclusion is necessary to allow for combining data from multiple API calls when processing larger areas. For instance, a subsequent API call could request data from $45^{\circ} \dots < 50^{\circ}$ latitude. Format used:

OpenMeteo : Téléchargement direct



Now, with the addition of the 9-kilometre ECMWF IFS model, the historical weather API provides access to a staggering 90 terabytes of meteorological data! Read the [blog article](#).

Location and Time

Location: ☒ Coordinates ☐ List ☐ Bounding box

Latitude North
69

Longitude West
-27

Longitude East
47

Latitude South
-24

Using bounding box will select grid cells within a specified area, excluding the top and right boundaries. For example, a latitude range of $40^{\circ} \dots < 45^{\circ}$ will not include any grid points on the 45° line. This exclusion is necessary to allow for combining data from multiple API calls when processing larger areas. For instance, a subsequent API call could request data from $45^{\circ} \dots < 50^{\circ}$ latitude. Format used:

```
&bounding_box=47,-85,47.5,-84.5
```

using (latitude1, longitude1, latitude2, longitude2) or (south, west, north, east).

The results are limited to a maximum of 1000 locations

The bounding box selection method is not recommended for building maps. Several projects are already leveraging Open-Meteo data for building maps; you can find an example [here](#).

Start date
2000-01-01

End date
2025-01-01

You can access past weather data dating back to 1940. However, there is a 5-day delay in the data. If you want information for the most recent days, you can use the [forecast API](#) and adjust the [Past Days](#) setting.

☐ Wind Speed (10 m)

2021 2022 2023 2024 2025

Hourly Weather Variables

☐ Temperature (2 m)

☐ Precipitation (rain + snow)

☐ Snow depth

☐ Weather code

☐ Sea Level Pressure

☐ Surface Pressure

☐ Cloud Cover Total

☐ Cloud Cover Low

☐ Cloud Cover Mid

☐ Cloud Cover High

☐ Reference Evapotranspiration (ET₀)

☐ Vapour Pressure Deficit

☐ Wind Speed (10 m)

☐ Wind Speed (100 m)

☐ Wind Direction (10 m)

☐ Wind Direction (100 m)

☐ Wind Gusts (10 m)

☐ Soil Temperature (0-7 cm)

☐ Soil Temperature (7-28 cm)

☐ Soil Temperature (28-100 cm)

☐ Soil Temperature (100-255 cm)

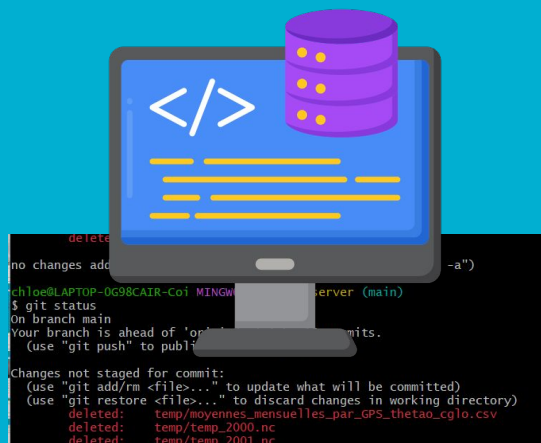
☐ Soil Moisture (0-7 cm)

☐ Soil Moisture (7-28 cm)

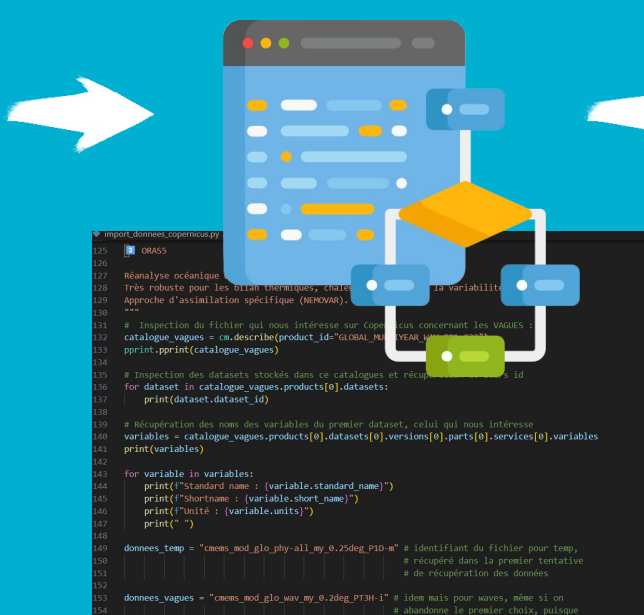
☐ Soil Moisture (28-100 cm)

☐ Soil Moisture (100-255 cm)

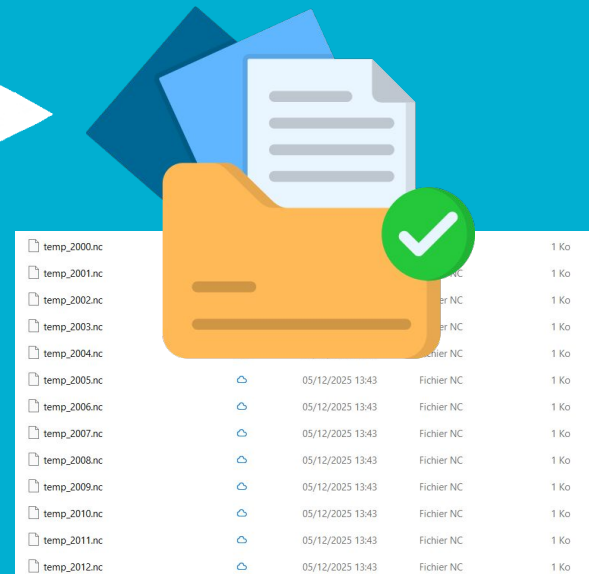
OpenMeteo : Programmation de la récupération via API



Git bash

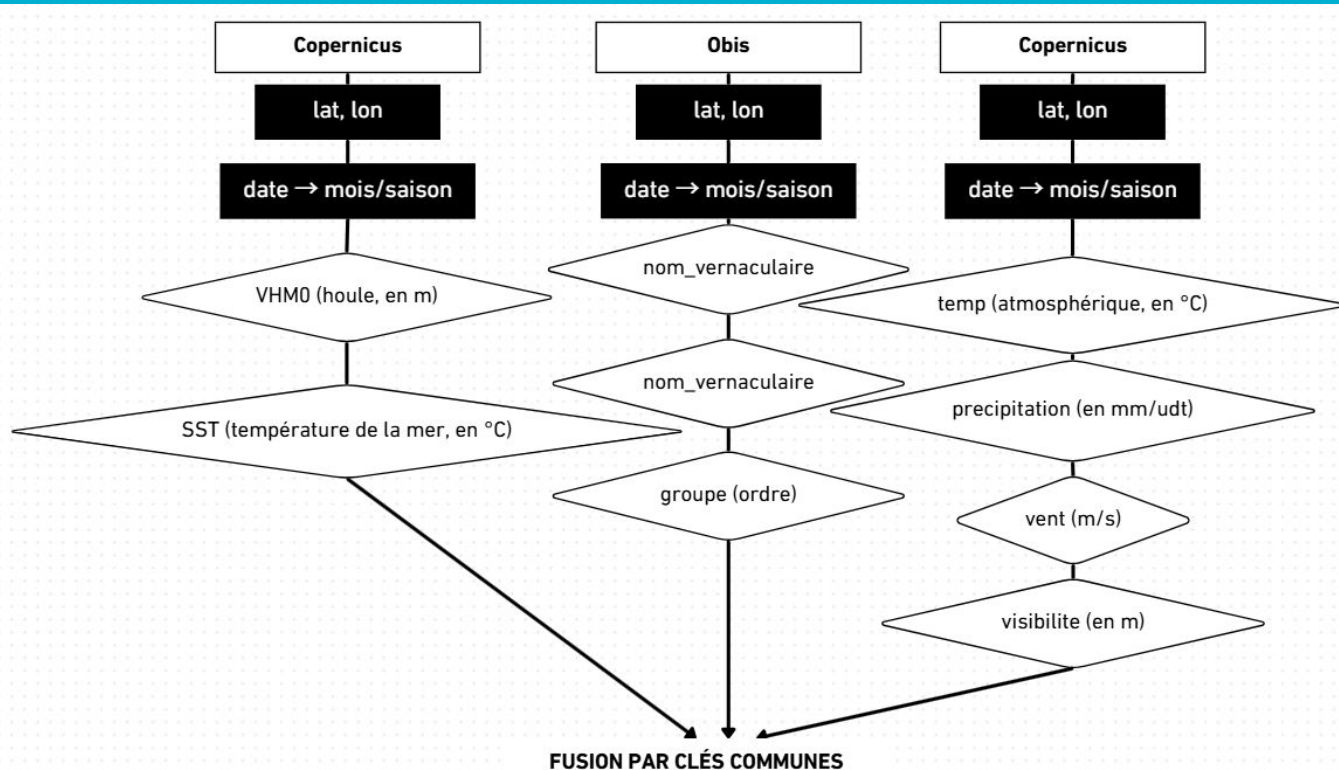


meteo.py



meteo.csv

Preprocess



Architecture

```
|— app.py
|— finalpoints/
|   └─ final_points.json      ← fichier généré
|— static/
|   └─ map_initialization.js  ← toute la logique front-end
|       └─ photos/           ← images des espèces
└─ templates/
    └─ index.html
```

- Structure simple volontaire

Carte interactive

- Jointure (json)
- [D3.js](#): contrôle total
et personnalisation
de l'affichage
- **assignRegion**(lat,lon)

```
"lat": 37.93126667,  
"lng": -24.54086667,  
"species": "Mesoplodon_bidens",  
"common_name": "Baleine à bec de Sowerby",  
"month": "june",  
"avg_temp": "19.72",  
"avg_rain": "0.72",  
"avg_wind": "26.36",
```

```
// --- 7. Logique de Filtrage ---  
function applyFilters() {  
  const groupFilter = document.getElementById('group-filter').value;  
  const regionFilter = document.getElementById('region-filter').value;  
  const monthFilter = document.getElementById('season-filter').value;  
  
  let filteredData = allObservations.filter(d => {  
    const matchGroup = groupFilter === 'all' || d.common_name === groupFilter;  
    const matchRegion = regionFilter === 'all' || d.region === regionFilter;  
    const matchMonth = monthFilter === 'all' || d.month === monthFilter;  
  
    return matchGroup && matchRegion && matchMonth;  
  });  
  
  document.getElementById('observation-count').textContent =  
    `${filteredData.length} ${filteredData.length > 1 ? 'observations trouvées' : 'obs'}`;  
  
  drawPoints(filteredData);  
}
```

regroupement (classe)

mois - saison

espèces - groupe d'espèces

Filtres de l'Observation

Groupe d'Espèces

Toutes les espèces (Nom Cc) ▾

Région Marine

Toutes les régions marines ▾

Période (Saison)

Toutes les périodes (Mois) ▾

32858 observations trouvées.

tempColorScale lie chaque point directement à une variable environnementale, ici la température de l'eau

Légende : Température de l'eau (°C)

5°C (Froid)

30°C (Chaud)



Ce qu'il nous reste à faire...

- Ajouter l'affichage des vagues
- Ajouter des conseils touristiques

Dauphin à bec étroit

Steno_bredanensis

15.88°C

Mois: january

Vent: 35.63 km/h | Pluie: 1.34 mm



Hauteur de Vagues (VHM0)

0.25 m

Conditions idéales (Classe D) - Mer calme à belle

Hauteur de Vagues (VHM0)

3.50 m

Conditions limitantes (Classes B/A) - Mer agitée à forte

Difficultés

- Bases de données volumineuses : git lfs
- Organisation du git
- Fr/eng (pas forcément possible installation de babel)

MERCI DE VOTRE ÉCOUTE LES COPAINS !

